tail is the longest part of the sperm and is slender and tapering. It is formed of a central axial filament or **axoneme** and an outer protoplasmic sheath. The lashing movements of the tail push the sperm forward. The human male ejaculates about 200 to 300 million sperms during coitus. It is estimated that around 60 percent of sperms must have normal shape of which at least 40 per cent must show vigorous motility for normal fertility.

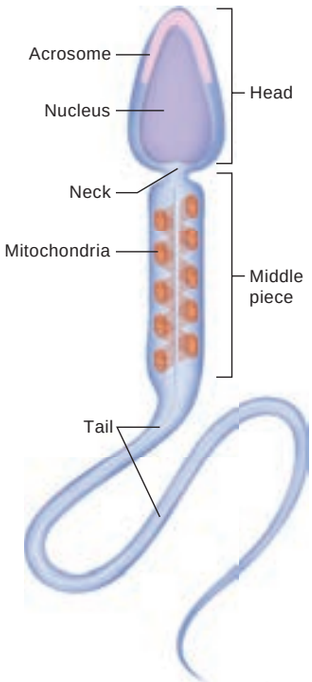


Fig. 2.7 Structure of human sperm



The sperm is the smallest human cell and the ovum or egg is the largest human cell.

Oogenesis

Oogenesis is the process of development of the female gamete or ovum or egg in the ovaries. During foetal development, certain cells in the germinal epithelium of the foetal ovary divide by mitosis and produce millions of **egg mother cells or oogonia**. No more oogonia are formed or added after birth. The oogonial cells start dividing and enter into Prophase I of meiotic division I to form the **primary oocytes** which are temporarily arrested at this stage. The primary oocytes then get surrounded by a single layer of granulosa cells to form the primordial or **primary follicles (Fig. 2.8 a)**. A large number of follicles degenerate during the period from birth to puberty, so at puberty only 60,000 to 80,000 follicles are left in each ovary.



Out of the million eggs women possess during birth, only about 300 to 400 will ovulate before menopause.

On the other hand, males produce more than 500 billion sperms in their life time.

The primary follicle gets surrounded by many layers of granulosa cells and a new theca layer to form the **secondary follicle**. A fluid filled space, the antrum develops in the follicle and gets transformed into a **tertiary follicle**. The theca layer gets organized into an inner theca interna and an outer theca externa. At this time, the primary oocyte within the tertiary follicle grows in size and completes its first meiotic division and forms the **secondary oocyte**. It is an unequal division resulting in the formation of a large haploid secondary oocyte and a first polar body. The first polar body disintegrates. During Fertilization, the secondary oocyte undergoes second meiotic division and produces a large cell, the **ovum** and a second polar body. The second polar body also degenerates. The tertiary follicle eventually becomes a mature follicle or **Graafian follicle**. If Fertilization does not take place, second meiotic division is never completed and the egg disintegrates. At the end of gametogenesis in females, each primary oocyte gives rise to only one haploid ovum.

Structure of ovum

Human ovum is **non-cleidoic, alecithal** and microscopic in nature. **(Fig. 2.8 (b))**. Its cytoplasm called **ooplasm** contains a large nucleus called the germinal vesicle. The ovum is surrounded by three coverings namely an inner thin transparent **vitelline membrane**, middle thick **zona pellucida** and outer thick coat of follicular cells called **corona radiata**. Between the vitelline membrane and zona pellucida is a narrow perivitelline space.

TN_GOVT_XII_Bio_Zoology_EM_chapter 2.indd 18 20-03-2020 10.32.01 AM